

REST API Learning Guide

This guide is designed for beginners who want to learn REST APIs for backend development using Java and Spring Boot.

1. Introduction to APIs

- What is an API?
- What is REST?
- REST Architecture
- Client-Server Communication

2. HTTP Fundamentals

- HTTP Methods (GET, POST, PUT, DELETE)
- HTTP Status Codes
- Headers and Body
- Request and Response

3. REST API Design

- Endpoints
- Resources
- Path Variables
- Query Parameters
- JSON Data Format

4. Building REST APIs with Spring Boot

- @RestController
- @RequestMapping
- @GetMapping
- @PostMapping
- @PutMapping
- @DeleteMapping

5. Database Integration

- Connecting MySQL
- Spring Data JPA
- CRUD Operations
- Repository Layer

6. Advanced REST API Concepts

- Validation
- Exception Handling
- DTOs
- Pagination and Sorting
- JWT Authentication

7. API Testing

- Testing with Postman
- Swagger Documentation
- Common API Errors
- Debugging APIs

8. Projects

- Student Management REST API
- Employee Management API
- E-Learning Backend API
- Authentication System API

Recommended Learning Timeline

Week	Focus Area
1	API Basics + HTTP
2	REST API Design
3	Spring Boot REST APIs
4	Database + CRUD APIs
5	Authentication + Projects

Recommended Tools:

- Spring Boot
- Postman

- Swagger
- MySQL Workbench
- IntelliJ IDEA / VS Code

REST APIs are essential for backend development and communication between frontend and backend systems. Mastering REST APIs will help you build scalable web applications and prepare for Java backend interviews.